

Exploring and Analysing Product Data Using Microsoft Excel

Submitted By:

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Course:

Program in AI Driven Data Analytics

1. Introduction

This project focuses on exploring and analysing product data using Microsoft excel as the primary analytical tool. The objective is to derive meaningful insights by Excel’s core functionalities, including formulas, logical functions, conditional formatting, and text manipulation. The analysis aims to demonstrate how Excel can be used for structured data exploration, pattern identification, and decision-support reporting in a business context.

2. Dataset Description

The dataset used for this analysis contains 10 attributes that capture product, temporal, and geographical details. The structure is as follows:

Column Name	Description
Product ID	Unique alphanumeric identifier for each of the product.
Day	Numeric Day of the month, parsed from Product ID.
Country Code	Standardized ISO- Style abbreviation of country Origin.
Month	Alphanumeric Month(Jan) parsed from the Product ID
Product Name	Descriptive commercial name of the product as listed for sale.
Brand Name	Brand associated with the product.
Price (\$)	Unit selling price of the Product in USD.
Quantity	Number of Units available or sold.
Category	High-level product grouping such as electronics, outdoor, etc. Used for category- wise segmentation.

Price Range	Categorical classification of the product based on its price, e.g., High price, Standard price.
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3. Functions Used in the Project

A. Basic Statistical Functions

1. SUM ()

Business Purpose: To compute the aggregate value of inventory by calculating the total sum of all product price. This provides a baseline metric for total product worth.

Syntax Applied: =SUM(D2:D35)

Result: 10,100

Insight: The combines listed price of all 34 product in the dataset is 10,100.

2. COUNTA()

Business Purpose: To determine the total number of records in the dataset by counting all non-blank entries in the Product Name column. This confirms dataset size and helps validate completeness.

Syntax Applied: =COUNTA(B2:B35)

Result: 34

Insight: The dataset contains 34 unique product entries available for analysis.

3. AVERAGE()

Business Purpose: To calculate the mean price point across all products. This metric helps identify the central pricing tendency and benchmark individual products.

Syntax Applied: =AVERAGE(D2:D35)

Result: \$297.06

Insight: The average product price is approximately \$297, indicating a mid-range product portfolio.

4. Minimum and Maximum Functions 1. MIN()

Business Purpose: To identify the lowest-priced product in the portfolio. This helps establish the entry-level price point and assess the lower bound of the product range.

Syntax Applied: =MIN(D1:D34)

Result: \$30

Insight: The minimum price of \$30 represents the most accessible product in the dataset. This value can be used as a benchmark for budget-category offerings. Result: 30

2. MAX()

Business Purpose: To identify the ceiling price of the product portfolio and define the premium offering.

Syntax Applied: =MAX(D2:D35)

Result: \$1,000

Insight: The highest-priced product is \$1,000. The total price range is \$970, calculated as \$1,000 - \$30. This indicates a wide portfolio spanning from budget to premium segments.

5. Logical Function

IF()

Business Purpose: To automate product classification based on price thresholds. This creates a new Price Range column that simplifies analysis by grouping products into actionable business segments.

Syntax Applied: =IF(D2>=500, "High Price", "Standard Price")

Result: Each product is labelled as either "High Price" or "Standard Price"

Insight: This logical test segments the portfolio into premium vs. standard offerings. The \$500 threshold was selected to isolate top-tier products for targeted marketing or margin analysis.

6. Conditional Functions

1. SUMIF()

Business Purpose: To calculate the total revenue potential of a specific product category. This isolates performance metrics for key business segments.

Syntax Applied: =SUMIF(F:F, "Electronics", D:D)

Result: \$8,050

Insight: Electronics products account for \$8,050 of the total portfolio value. Compared to the overall SUM() of \$10,100, Electronics represents ~79.7% of total value, indicating it is the dominant category.

2. COUNTIF()

Business Purpose: To quantify the number of products within a specific price band. This helps assess the density of budget-tier offerings.

Syntax Applied: =COUNTIF(D:D, "<100")

Result: 11

Insight: This count reveals how many products are priced under \$100. A high count would suggest a strategy focused on total sales.

7. Text Functions

1. LEFT()

Business Purpose: To extract the day component from the beginning of the Product ID. This enables day-level trend analysis and supports time-series reporting.

Syntax Applied: =LEFT(A2, 2)

Result: 28

Insight: Successfully isolated "28" as the day value. This allows for analysis of sales by specific days of the month.

2. RIGHT()

Business Purpose: To extract the country code from the end of the Product ID. This creates a dedicated field for regional segmentation and market-based analysis.

Syntax Applied: =RIGHT(A2,2)

Result: US

Insight: Identified "US" as the market code. This derived column enables filtering and aggregation by geography without manual data entry.

3. MID()

Business Purpose: To extract the month component from the middle of the Product ID. This is critical for identifying seasonality and performing month-over-month performance comparisons.

Syntax Applied: =MID(A2,4,3)

Result: JAN

Insight: Extracted "JAN" as the month value. This function completes the deconstruction of Product ID into three distinct, analysis-ready attributes.

8. Conclusion

This project successfully demonstrated the application of Microsoft Excel as a foundational tool for structured data analysis within a business context. By employing a combination of statistical functions, logical tests, conditional aggregations, and text manipulation, raw product data was transformed into actionable insights.

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